# Model Beta B07 Failure Analysis Research Brief

HELIX FORGE LAB / p04-full-006138 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded failure analysis evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000034 (research/programs/model-beta/experiments/configs/2026-07-13/model-beta-b07/comparison-study/tables/model-beta-b07-comparison-stu

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 194 | 0.891 | 0.853 | 0.872 |
| slice-02 | 214 | 0.873 | 0.866 | 0.869 |
| slice-03 | 242 | 0.890 | 0.879 | 0.884 |
| slice-04 | 180 | 0.872 | 0.855 | 0.863 |
| slice-05 | 215 | 0.889 | 0.868 | 0.878 |